

Davide BRESSAN

PhD Candidate | Bioinformatician/Computational biologist

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I am a Post doc researcher in the Chiacchiera Lab at the Department of Cellular, Computational and Integrative Biology, University of Trento, Italy. Trained as a computer scientist and bioinformatician, I now work as a computational biologist in cancer research. My research is focused on the epigenetics of liver regeneration and tumorigenesis.

WORK EXPERIENCE

Current February 2025	Postdoctoral researcher, UNIVERSITY OF TRENTO, Italy ➢ Computational biologist in the group of Prof. Fulvio Chiacchiera (Laboratory of Stem Cells and Cancer Genomics). RNA-SeqChIP-SeqGeneticsCancerR
January 2025 November 2020	PhD in Biomolecular Sciences, UNIVERSITY OF TRENTO, Italy ➢ Epigenetic regulation of liver homeostasis and carcinogenesis. Group of Prof. Fulvio Chiacchiera. ➢ Transdisciplinary program in Computational Biology RNA-SeqChIP-SeqGeneticsCancerR
January 2024 November 2022	Teaching assistant, course of Scientific Programming for Data Science, UNIVERSITY OF TRENTO, Italy ➢ Preparation and delivery of computer laboratory lectures for the Master in Data Science. ➢ Exam preparation PythonData structuresAlgorithmsStudent supervision
July 2020 March 2020	Intern Bioinformatician, WINCLOVE PROBIOTICS B.V, Amsterdam, Netherlands ➢ Trainee in the company's research and innovation team. ➢ Analysis of metagenomic data from patients with inflammatory bowel disease using machine learning algorithms MetagenomicsMachine learningPython
July 2020 March 2020	Teaching assistant, course of Fundamentals of Bioinformatics, VRIJE UNIVERSITEIT, Amsterdam, Netherlands ➢ Guidance and evaluation of students in the first year of the Master in Bioinformatics and System Biology PythonStudent supervision
February 2020 August 2019	Intern Bioinformatician, NETHERLANDS CANCER INSTITUTE (NKI), Amsterdam, Netherlands ➢ Internship as a bioinformatician in the group of Prof. Daniel Peeper (Department of Molecular Oncology and Immunology) ➢ Development of an R program for the analysis of splice variants. ➢ Thesis title: "The role of transcript variants in immunotherapy resistant melanoma". RNA-SeqRMelanoma

FORMATION

2020-2025	Ph.D in Biomolecular Sciences <i>cum laude</i> - University of Trento - Italy
2018-2020	Master of Science in Bioinformatics and System Biology – Universiteit van Amsterdam - Vrije Universiteit Amsterdam - Netherlands
2015-2018	Bachelor's Degree in Bioinformatics – University of Verona - Italy

LANGUAGE

Italian ● ● ● ● ●
English ● ● ● ● ○
Dutch ● ○ ○ ○ ○

TECHNICAL SKILLS

- Programming Languages: R, Python, Java
- Data analysis: Bulk and sc RNA-seq, ChIP-seq, ATAC-seq, Genomics
- Data Visualization: ggplot2, matplotlib
- Bash and pipeline development (Snakemake)

PUBLICATIONS

MODELLING SPORADIC PROGRESSIVE SUPRANUCLEAR PALSY IN 3D MIDBRAIN ORGANOID: RECAPITULATING DISEASE FEATURES FOR IN VITRO DIAGNOSIS AND DRUG DISCOVERY

Article in Annals of Neurology - January 2025 -DOI: <https://doi.org/10.1002/ana.27172>

Co-author

scRNA-seq Neuroscience

RAR γ -FOXA1 SIGNALING PROMOTES LUMINAL IDENTITY IN PROSTATE PROGENITORS AND IS DISRUPTED IN PROSTATE CANCER

Article in EMBO Reports - December 2024 -DOI: <https://doi.org/10.1038/s44319-024-00335-y>

Co-first author

Ar Foxa1 Prostate Cancer Organoids

SPIKEFLOW: AUTOMATED AND FLEXIBLE ANALYSIS OF CHIP-SEQ DATA WITH SPIKE-IN CONTROL

Article in NAR Genomics and Bioinformatics - August 2024 - DOI: <https://doi.org/10.1093/nargab/lqae118>

First author

ChIP-seq Spike-in Pipeline Bioinformatics

INCREASED GENOMIC INSTABILITY AND RESHAPING OF TISSUE MICROENVIRONMENT UNDERLIE ONCOGENIC PROPERTIES OF ARID1A MUTATIONS

Article in Science Advances - March 2024 -DOI: 10.1126/sciadv.adh4435

Co-first author

Arid1A Liver Cancer Mutagenesis

POSTERS AND TALKS

MSWI/SNF MUTATIONS AFFECT LIVER-SPECIFIC REGENERATIVE RESPONSE AND RESHAPE TISSUE MICROENVIRONMENT

SIBBM 2024 - Frontiers in Molecular Biology

June 2024, Trento, Italy

Poster

RSG-ITALY YOUNG SYMPOSIUM

CIBB 2023 - 18TH CONFERENCE ON COMPUTATIONAL INTELLIGENCE METHODS FOR BIOINFORMATICS & BIOSTATISTICS.

September 6-8 2023, Padova, Italy

Session Chair and Program Committee

INCREASED GENOMIC INSTABILITY AND RESHAPING OF TISSUE MICROENVIRONMENT DRIVE ARID1A-DEPENDENT CANCER FORMATION

SIBBM 2023 - Frontiers in Molecular Biology

June 2023, Bari, Italy

Oral Talk

THE ROLE OF MSWI/SNF COMPLEX IN PRESERVING LIVER TISSUE HOMEOSTASIS AND PREVENTING HEPATOCELLULAR CARCINOMA FORMATION

The 33° PEZCOLLER SYMPOSIUM - Presenting author

June 2022, Trento, Italy

Poster

ROLE OF THE MSWI/SNF CHROMATIN REMODELING COMPLEX IN PRESERVING LIVER HOMEOSTASIS

EACR conference - Presenting author

June 2022, Seville, Spain

Poster

“ REFERENCES

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